

# Project ATLAS-MORPH

## Sovereign Diacritic-Aware Tokenization & Low-Rank Inference Acceleration for N-ATLaS

**Prepared for:** Mrs. Hauwa Ibrahim Aminu

**Institution:** Nile University of Nigeria (Abuja)

**Track:** Academia & Research Track (Problem Statement 01)

**Deadline:** 12 October 2026, 23:59 WAT

### 1. EXECUTIVE SUMMARY

**ATLAS-MORPH** is an academic research and developer tooling project engineered by student researchers from the **Department of Computer Science, Nile University of Nigeria**, under the faculty supervision of **Mrs. Hauwa Ibrahim Aminu**.

The project directly resolves a fundamental architectural bottleneck in **N-ATLaS** (Nigeria's sovereign 8B multilingual LLM): the severe **3.8x "Tokenization Tax" and memory latency penalty** experienced when processing tonal and diacritic African languages (Yorùbá, Hausa, and Igbo). By deploying an **orthographic Unicode normalizer** paired with a **4-bit low-rank KV-cache kernel**, ATLAS-MORPH reduces token fertility from **3.82 down to ~1.4 tokens/word** and cuts active memory from **16.2 GB to ~5.2 GB VRAM**, enabling N-ATLaS to run at **>2.5x speed** on standard student laptops and university lab workstations.

### 2. TECHNICAL INNOVATION & ARCHITECTURE PIPELINE

#### STAGE 1: Unicode Normalizer

Canonical NFC/NFD reconciliation merges fragmented tonal accents (ọ, ẹ, à, â, ã) into virtual subwords, eliminating raw byte fallback.

#### STAGE 2: 4-Bit KV-Cache

AutoAWQ & Paged KV-Cache tailored for N-ATLaS's 32-layer GQA, slashing memory footprint from 16.2GB to 5.2GB for budget GPUs.

#### STAGE 3: 1-Line Drop-in SDK

import atlas\_morph as am wrapper  
+ interactive web dashboard providing live side-by-side latency and VRAM telemetry.

#### Standard Llama-3 / N-ATLaS Tokenizer:

Yorùbá & Hausa Fertility: **3.82 tokens/word** (320% compute penalty).

#### With ATLAS-MORPH Acceleration:

Optimized Fertility: **~1.4 tokens/word** (>2.5x throughput acceleration).

### 3. INSTITUTIONAL & ACADEMIC BENEFITS FOR NILE UNIVERSITY

Securing a top placement in the NAIC 2026 Academia Track delivers major institutional prestige:

| Award Category            | Impact & Recognition for Nile University & Faculty                                        |
|---------------------------|-------------------------------------------------------------------------------------------|
| N-ATLaS Co-Contributor    | Official co-credit on the sovereign model card and integration into N-ATLaS v2.           |
| NCAIR Lab Partnership     | Nile University becomes an official research partner institution with NCAIR.              |
| National AI Registry      | Official induction of supervisor and researchers into the Federal Ministry's AI Registry. |
| Research & Compute Grants | Federal sponsorship for compute credits, academic publishing, and cash prizes.            |

### 4. MANDATORY PORTAL SUBMISSION DELIVERABLES (OCTOBER 12)

| #  | Portal Requirement | Format | Deliverable Description                                                      |
|----|--------------------|--------|------------------------------------------------------------------------------|
| 01 | Working Artefact   | URL    | Open-source Python package (atlas-morph) on GitHub with setup documentation. |

|    |                                     |     |                                                                                     |
|----|-------------------------------------|-----|-------------------------------------------------------------------------------------|
| 02 | <b>N-ATLAS Integration Evidence</b> | ZIP | Execution logs and scripts demonstrating direct inference on NCAIR1/N-ATLaS.        |
| 03 | <b>Real-World Validation</b>        | PDF | Empirical benchmark report (>2.5x speedup) + feedback from 2+ external lab testers. |
| 04 | <b>Technical Documentation</b>      | PDF | Formal architecture report detailing normalizer and 4-bit KV-cache mechanics.       |
| 05 | <b>Video Demonstration</b>          | URL | 3–5 minute video demonstration showcasing live token acceleration end-to-end.       |
| 06 | <b>Team Profile</b>                 | PDF | Academic bios, affiliations, and role descriptions for all team members.            |
| 07 | <b>HOD Endorsement Letter</b>       | PDF | 1-page endorsement letter signed by the HOD on Nile University letterhead.          |

## 5. OFFICIAL TEAM ROSTER

**FACULTY SUPERVISOR**

**Mrs. Hauwa Ibrahim Aminu**

Lecturer, Dept. of Computer Science, Nile University

**TEAM LEAD & SYSTEMS ARCHITECT**

**Medugu Wali**

Dept. of Computer Science, Nile University

**NLP & TESTING SPECIALIST**

**Mutmainnah Magaji**

Dept. of Computer Science, Nile University

**FULL-STACK & BENCHMARKING LEAD**

**Ojo Timothy**

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